SEDAR

Southeast Data, Assessment, and Review

SEDAR 94

Florida Hogfish

SECTION IV: Research Recommendations

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1. DATA WORKSHOP RESEARCH RECOMMENDATIONS

1.1 LIFE HISTORY RESEARCH RECOMMENDATIONS

1.1.1 Stock Definition

The LHWG recommended expanding genetic sampling in the currently defined unit stocks for Hogfish to reinforce or challenge respective boundaries. In the FLK/EFL stock, sampling does not exist from central (e.g., Cape Canaveral area) and northeastern Florida; investigating this could provide insight into the assumed connection with southeastern Florida as well as any potential source and sink dynamics. Furthermore, no Hogfish have been sampled in GA or SC; any fish here are currently assumed to be defined with the stock sampled off NC, but uncertainty in that boundary exists and whether portions may be further spillover from south Florida. The group also recommends the use of updated genetic methods (e.g., RADseq derived genomic data, SNPS; see Wallace et al. *in prep*. for Spotted Seatrout in Florida) to better inform these biological and managerial boundaries. Lastly, there is no genetic data published from Hogfish in the Caribbean, but preliminary results from mitochondrial DNA samples in Puerto Rico and St. Thomas indicate some degree of genetic connectivity exists with those from the WFS and southeast FL (A. Janosik, *pers. comm*. 5/15/25).

1.1.2 Age and Growth

The LHWG expressed the need to increase the annual fishery-dependent and -independent age sampling of Hogfish. Current sample sizes make detecting and tracking year classes difficult or inadequate in most years without the reliance on data from life history studies. However, the LHWG recognizes the challenge of this as the fisheries tend to be more opportunistic for Hogfish. For the FLK/EFL stock, sampling also needs to expand to include more areas of refuge (e.g., marine no-take reserves; Muñoz et al. 2010) and areas currently unsampled (e.g., northeast FL) to better inform population demographics. Hogfish in both the heavily fished areas of the FLK/EFL and the nearshore areas of the WFS where fishing predominantly occurs exhibit truncated age-at-length data but can grow larger and live longer when able to escape to refuge.

1.1.3 Natural Mortality

The field of natural mortality is not yet in a position to establish 'best practices' (Maunder et al. 2023) and suggested 'good practices' are often trade-offs between reliability and availability of the data. More direct methods of estimating mortality, such as mark-recapture or acoustic telemetry tagging, are generally recommended over empirical methods but have largely been unavailable for Hogfish until recently. Ellis et al. (2025) conducted a pilot study to estimate discard mortality and successfully tagged 42 Hogfish (including 38 tagged with acoustic tags) from 45 trips on the WFS between May 2019 and February 2025, showing the viability for a larger scale study. Effort into acoustic telemetry tagging requires a large enough array of detectors to minimize incomplete detections, but the high site fidelity of Hogfish seem to make it a good candidate as movement out of the array area for extended periods (i.e., are the change in numbers due to mortality or migration) may be minimized. Ellis et al. (2025) also reported 3 re-captured individuals, but none were reported by recreational anglers; rather, they were determined via acoustic detection which alleviates human reporting issues common in conventional tagging.

1.1.4 Reproduction

The LHWG commends the amount of histological data that has been available for Hogfish from life history research studies but also notes that all reproductive data used here were at least 18 years old. If the original histology slides are still available, it is recommended to re-analyze them

following current best practices (Lowerre-Barbieri et al. 2022, Lowerre-Barbieri and Friess 2023) which would also allow comparison with any reproductive data collected in the future for determining size- and age-at-maturity and spawning seasonality. The LHWG also notes the impact of top-down size selective fishing mortality on harem-disruption, social breakdown, reproduction, and possible sperm limitation for Hogfish and recommends additional *in situ* sampling (e.g., diver surveys; Muñoz et al. 2010) to help evaluate recovery of these from the impacts of fishing.

1.2 COMMERCIAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

Landings

- Improve gear, area, and effort data reported for each trip on state Trip Ticket programs to ensure more accurate data for future assessments.

Discards

- Continue to research alternative methods and analyses for estimating total effort during the closed season of a fishery.

- Recommend the observer program investigate the allocation of some observer coverage for focused trips to aid in future SEDARs as a subset of existing strata.
    - For Hogfish, this may include allocating addition effort to increase the probability of Hogfish encounters in areas of higher population density (SE FL, FL Keys, West Florida)

Biosampling

- Hold more frequent TIP trainings and increase port agent buy-in through information exchange with assessment and SEDAR staff focused on the usage of data collected by TIP staff to continue to refine data quality.
- Increase the priority of Hogfish sampling throughout its' distribution (potentially allocating funds to purchase landed Hogfish to increase the number of aged fish)

1.3 RECREATIONAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

1.3.1 Research Recommendations

- Additional release mortality research specially investigating mortality of surface released fish and/or vented fish caught by hook and line in the recreational fishery. This should include research into the incidence of barotrauma, its causes, and its relationship to release mortality.

- Improve precision of MRIP and SRFS estimates for all modes through increased sampling
- Additional age data collection from all recreational sources by region (WFS, FLK/Keys), by gear (spear, hook and line), and by fishing area
- Increasing sampling to intercept and collect biological information from private recreational Hogfish anglers
- Explore methods to acquire catch information from private boat anglers returning to private docks in addition to public access points

1.3.2 Other Recommendations

Issue: Providing a potential model start year recommendation according to recreational data availability and uncertainty. The RWG discussed a potential recommendation to start the assessment model in 1986, as has been done in the previous Hogfish assessment (SEDAR 37) and other SEDAR stock assessments for reef fishes.

Recommendation: Catch estimates from the early years of the MRIP survey (e.g., 1981-1985) are highly variable and tend to result in higher CVs than those estimated in subsequent years. Additionally, some early years (1984-1985) in the WFS have zero estimated landings. The group has no particular concerns with retaining and using the recreational data in these early years as inputs into the assessment model but recognizes the above reasons could be used for justification of removal. The RWG concluded that both options are valid and supported and recommends this decision be made during the assessment process based on other aspects of data availability and modeling needs.

1.4 MEASURES OF POPULTAION ABUNDANCE RESEARCH RECOMMENDATIONS

- **Evaluate the Integration of Supplemental SRFS Intercept Data:** The IWG recommends investigating the feasibility and utility of incorporating supplemental SRFS intercept data into MRIP indices of abundance to improve precision.
- **Expand Length Data Collection through REEF/SMILE:** Continue collecting additional length-frequency data via the SMILE project for consideration in future stock assessments.

- **Collect Genetic Data to Assess Stock Mixing:** Obtain genetic samples from broad-scale fishery-independent surveys (e.g., SEAMAP trawl) that encompass both the West Florida Shelf (WFS) and the Keys-East Florida (Keys-EFL) stocks to evaluate mixing rates between the two.
- **Assess Environmental Stressors:** Investigate the impacts of seagrass habitat decline in the Indian River Lagoon and the effects of red tide events on Hogfish population dynamics and distribution.
- **Develop a Juvenile Abundance Survey in the Keys-EFL Region:** Design and implement a fishery-independent survey targeting juvenile Hogfish to generate indices of abundance for the Keys-EFL stock, addressing a current data gap in early life history stages.
- **Examine Migration Patterns:** Study the seasonal and spatial timing of inshore and offshore migrations to better understand life history patterns and inform management strategies.

2. ASSESSMENT PROCESS RESEARCH RECOMMENDATIONS

No consolidated research recommendations section was included in the assessment report.

3. REVIEW PANEL RESEARCH RECOMMENDATIONS

The Review Panel supports the research recommendations developed during the Data and Assessment Workshops and identified several additional priorities that could improve future hogfish assessments.

1. Improve understanding and modeling of protogynous hermaphroditic population dynamics.
    - This represents a high research priority. Future work should investigate the recruitment and reproductive dynamics of protogynous, harem-forming species and evaluate how social structure affects sex transition. The current assessment represents sex transition primarily as an age-dependent process, whereas hogfish sex change may also depend strongly on the local abundance or density of males. Research should evaluate whether the probability of female-to-male transition changes with male density and how fishing-induced changes in population sex ratio affect reproductive output. Simulation studies and explicitly sex-structured population models could be particularly useful for evaluating these processes and the implications of size-selective harvest regulations for protogynous species.

- Related work should investigate the potential for sperm limitation or male limitation and the prevalence of skip spawning. These processes could influence the relationship between conventional measures of spawning biomass and realized reproductive output and may help explain why a conventional stock-recruit relationship is difficult to estimate for this species.

2. Improve recreational spearfishing information.

   - Uncertainty in recreational spear landings remains one of the more important data limitations for the WFS assessment. Existing intercept designs may inadequately sample portions of the spearfishing fleet, particularly fishers operating from private access points. Targeted survey designs should be considered to evaluate the magnitude and potential bias of recreational spear removals and to determine whether existing CPUE observations adequately represent the broader spearfishing population. Improvements in this area would increase confidence in historical fishing pressure, current stock status, and sustainable harvest reference points.

3. Evaluate fishery-independent survey coverage and data gaps.

   - Available fishery-independent indices, including those excluded during the DW, should be systematically evaluated with respect to geographic coverage of hogfish habitat, sample size, and availability of age and length information. This could identify important spatial or temporal gaps and help prioritize future monitoring.

4. Evaluate alternative assessment and uncertainty methods.

   - Future methodological work should compare uncertainty estimates derived from MCMC, parametric bootstrap, and asymptotic normal approximations. Comparisons on both normal and log scales could help determine circumstances in which conventional asymptotic approximations provide sufficiently reliable inference and when more computationally intensive MCMC or bootstrap procedures are warranted.

   - Alternative MCMC algorithms, including Hamiltonian Monte Carlo and related approaches, should be investigated where feasible to determine whether they improve convergence and posterior exploration relative to the current Metropolis-Hastings implementation.

   - Additional consideration should also be given to assessment frameworks capable of incorporating random effects or state-space processes, including time-varying fishing mortality where appropriate.

5. Evaluate Stock Synthesis implementation choices that are not well understood.

   - Additional research should examine the statistical properties and consequences of using super-periods to address sparse composition data. Sensitivity analyses should also evaluate the specification of growth platoons, given the limited experience of many assessment practitioners and reviewers with this feature.

   - The implications of removing the recruitment-deviation sum-to-zero constraint for MCMC proxy models should also be examined further. The current workaround

was considered acceptable for the WFS assessment, but a better understanding of the statistical consequences would benefit future applications.

6. Investigate sex-specific growth.
    - Although residual diagnostics supported the assumption of equal growth rates for males and females in the current assessment, additional biological and modeling work should evaluate whether sex-specific growth differences are important enough to affect future assessment results.
7. Evaluate environmental effects on hogfish population dynamics.
    - Future research should consider the effects of environmental disturbances and stressors, including red tides and other harmful algal blooms, cold events, and oil spills. Potential environmental effects on natural mortality, recruitment, movement, and habitat use could be important sources of temporal variability not explicitly represented in the current assessment.
8. Develop stochastic projections that propagate assessment uncertainty.
    - The MCMC framework provides an opportunity to propagate parameter uncertainty into future projections. Stochastic projections could quantify probabilities of SSB falling below biomass thresholds or fishing mortality exceeding applicable reference points under alternative management scenarios. Incorporating such approaches more routinely into the SEDAR process could improve communication of management risk and make fuller use of the uncertainty analyses already conducted as part of the assessment.

- **Provide recommendations on possible ways to improve the SEDAR process**

  The Review Panel commented that some TORs may need to be revisited or clarified. For example, the text on BSIA may need updating, and the evaluation of the projection results may be a task for the SSC, rather than a Review Panel. Also, the Review Panel noted that the bridging analysis (see TORs for the assessment phase) was not very informative given the length between assessments and resulting changes in the data and the model. It was therefore not evaluated by the Review Panel.